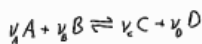
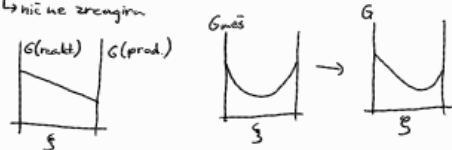


$\Delta_r G < 0$ - spontan proces

reaktanti $\rightleftharpoons$ produkti

ξ - obseg reakcije

$0 \leq \xi \leq 1$
 $\rightarrow$ vse zreagira
 $\rightarrow$ nič ne zreagira

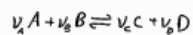


$\vec{v} = k[A][B]$ ali: $v_i = k_i[A][B]$
 ni vektor, ampak nakazuje potek reakcije v desno

$$v_i = k_{-i}[C][D]$$

v ravnotežju: $v_i = v_{-i}$

$$\Delta G = \sum \mu_i dn_i = \begin{cases} < 0 & \text{spontan proces} \\ = 0 & \text{ravnotežje} \\ > 0 & \text{ne poteče spontan} \end{cases}$$



$n_A^*, n_B^*, \nu_C^*, n_D^* \dots$ zač. množila snovi

$$\mu_i = \mu_i^* + RT \ln a_i$$

$$n_A = n_A^* - \nu_A \xi \quad dn_A = -\nu_A d\xi$$

$$n_B = n_B^* - \nu_B \xi \quad dn_B = -\nu_B d\xi$$

$$n_C = n_C^* + \nu_C \xi \quad dn_C = \nu_C d\xi$$

$$n_D = n_D^* + \nu_D \xi \quad dn_D = \nu_D d\xi$$

$$dG = \mu_A dn_A + \mu_B dn_B + \mu_C dn_C + \mu_D dn_D$$

$$= (-\nu_A \mu_A - \nu_B \mu_B + \nu_C \mu_C + \nu_D \mu_D) d\xi$$

$$\left(\frac{\partial G}{\partial \xi} \right) = \Delta_r G = \mu_C \nu_C + \mu_D \nu_D - \mu_A \nu_A - \mu_B \nu_B =$$

$$= \nu_C (\mu_C^* + RT \ln a_C) + \dots =$$

$$= \nu_C \mu_C^* + \nu_D \mu_D^* - \nu_A \mu_A^* - \nu_B \mu_B^* + RT \ln \frac{a_C^{\nu_C} a_D^{\nu_D}}{a_A^{\nu_A} a_B^{\nu_B}} =$$

$$\Delta_r G = \Delta_r G^* + RT \ln \frac{a_C^{\nu_C} a_D^{\nu_D}}{a_A^{\nu_A} a_B^{\nu_B}}$$

Q - reakcijski kvocient

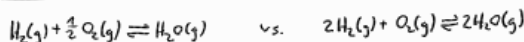
$$a_i = \frac{f_i}{p^*} \quad \text{ali} \quad a_i = \gamma_i \frac{c_i}{c^*}$$

(plin) (raztopina)

ravnotežje: $\Delta_r G = 0 \rightarrow Q = K_a$
 $\rightarrow$ ravnotežna konstanta

$$RT \ln K_a = -\Delta_r G^*$$

$$\Delta_r G = -RT \ln K_a + RT \ln Q = RT \ln \frac{Q}{K_a}$$



$$K_1 = \frac{a(\text{H}_2\text{O})}{a(\text{H}_2) a^{1/2}(\text{O}_2)}$$

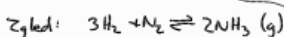
$$K_2 = \frac{a(\text{H}_2\text{O})^2}{a(\text{H}_2)^2 a(\text{O}_2)} = K_1^2$$

$$a_i = \frac{f_i}{p^*}, \quad f_i = \gamma_i p_i$$

$\rightarrow$ aktivnostni koeficient

$$K_a = \frac{\left(\frac{f_C}{p^*} \right)^{\nu_C} \left(\frac{f_D}{p^*} \right)^{\nu_D}}{\left(\frac{f_A}{p^*} \right)^{\nu_A} \left(\frac{f_B}{p^*} \right)^{\nu_B}} = \underbrace{\gamma_C^{\nu_C} \gamma_D^{\nu_D}}_{K_f} \underbrace{\left(\frac{p_0}{p^*} \right)^{\nu_C} \left(\frac{p_0}{p^*} \right)^{\nu_D}}_{K_p}$$

$$K_f = \frac{\left(\frac{f_3}{f^*} \right)^{\nu_3} \left(\frac{f_4}{f^*} \right)^{\nu_4}}{\left(\frac{f_1}{f^*} \right)^{\nu_1} \left(\frac{f_2}{f^*} \right)^{\nu_2}}$$



$p[\text{bar}]$	K_f	K_f	K_f
10	6,53	0,335	6,56
100	7,25	0,88	6,38
600	12,94	0,477	6,34

$K_a = K_f$ za pline

$$K_a = f(T)$$

$$K_f, K_p = f(p, T)$$

$$p = \frac{n}{V} RT = c RT, \quad c = \frac{p}{RT}$$

$$K_p = \frac{\left(\frac{c_A RT}{p^*}\right)^{v_A} \left(\frac{c_B RT}{p^*}\right)^{v_B} \left(\frac{c_C^* RT}{p^*}\right)^{v_C}}{\left(\frac{c_A RT}{p^*}\right)^{v_A} \left(\frac{c_B RT}{p^*}\right)^{v_B} \left(\frac{c_C^* RT}{p^*}\right)^{v_C}} = \underbrace{\left(\frac{c_A}{c^*}\right)^{v_A} \left(\frac{c_B}{c^*}\right)^{v_B}}_{K_c} \left(\frac{RT}{p^*}\right)^{v_C + v_B - v_A - v_B}$$

$$\rightarrow K_p = K_c \left(\frac{RT}{p^*}\right)^{\Delta v}$$

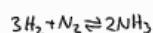
PLINI:

$$p_i = x_i p$$

$$K_x = \frac{x_C^{v_C} x_D^{v_D}}{x_A^{v_A} x_B^{v_B}}$$

$$K_p = K_x \left(\frac{p}{p^*}\right)^{\Delta v}$$

Primer: sinteza NH_3



$$T = 400\text{K}$$

$$n(\text{H}_2) = 0,75\text{mol}$$

$$n(\text{N}_2) = 0,25\text{mol}$$

$$K_p = 35$$

a) $p = 1\text{bar}$
 $V = ?$

b) $V = 3,26\text{L}$
 $p = ?$

Vpliv p na ravnotežje

$$K_x = K_p \left(\frac{p}{p^*}\right)^{\Delta v} / \left(\ln \left(\frac{\partial}{\partial p}\right)\right)_T$$

$$\left(\frac{\partial \ln K_x}{\partial p}\right)_T = \left(\frac{\partial \ln K_p}{\partial p}\right)_T - \Delta v \frac{1}{p}$$

= 0 za idealne pline

b) $p = \frac{n}{V} RT = 6,78\text{kPa}$

$$x(\text{H}_2) = \frac{n(\text{H}_2)}{n} = \frac{0,75-3\alpha}{1-2\alpha}$$

$$x(\text{N}_2) = \frac{0,25-\alpha}{1-2\alpha}$$

$$x(\text{NH}_3) = \frac{2\alpha}{1-2\alpha}$$

$$\rightarrow K_x = \frac{x(\text{NH}_3)^{2v_{\text{NH}_3}}}{x(\text{N}_2)^{v_{\text{N}_2}} x(\text{H}_2)^{v_{\text{H}_2}}} = \frac{\left(\frac{2\alpha}{1-2\alpha}\right)^2}{\frac{0,25-\alpha}{1-2\alpha} \left(\frac{0,75-3\alpha}{1-2\alpha}\right)^3} = \frac{(2\alpha)^2 (1-2\alpha)^2}{(0,25-\alpha)(0,75-3\alpha)^3} = K_p \left(\frac{p^*}{p}\right)^{\Delta v} = 35$$

$$\alpha_1 = 0,1652$$

$$\alpha_2 = 0,334 \text{ NI OK}$$

$$\alpha_{1,2} \in \mathbb{R}$$

$$V = \frac{n}{p} RT = 0,02227\text{m}^3$$

Vpliv T na ravnotežje

$$\Delta_r G^* = -RT \ln K_x$$

$$\ln K_x = -\frac{\Delta_r G^*}{RT} / \left(\frac{\partial}{\partial T}\right)_p$$

$$\left(\frac{\partial \ln K_x}{\partial T}\right)_p = -\frac{1}{R} \left(\frac{\partial \Delta_r G^*/T}{\partial T}\right)_p = -\frac{1}{R} \left(-\frac{\Delta_r H^*}{T^2}\right) = \frac{\Delta_r H^*}{RT^2}$$

Le Chatelier-ov vpliv:

$\Delta v = 0 \rightarrow$ tlak ne vpliva na ravnotežje

$\Delta v > 0 \rightarrow p \uparrow K_x \downarrow$

$\Delta v < 0 \rightarrow p \uparrow K_x \uparrow$

Le Chatelier-ov vpliv:

$\Delta_r H^* > 0$, endotermna $T \uparrow \rightarrow K_x \uparrow$

$\Delta_r H^* < 0$, ekstermna $T \uparrow \rightarrow K_x \downarrow$

Naloga: $\text{N}_2\text{O}_4(g) \rightleftharpoons 2\text{NO}_2(g)$, $T = 318\text{K}$, $p = 1\text{bar}$, $\alpha = 0,38$
 $\rightarrow$ določiti razpadlih molekul N_2O_4

$$\frac{p}{p^*} = 10\text{bar}$$

$$\alpha_2 = ?$$

$$\Delta v = 1, p = p^*$$

	N_2O_4	NO_2
začetek	n_0	0
zreagira	αn_0	$2\alpha n_0$
v ravnotežju	$n_0(1-\alpha)$	$2\alpha n_0$

$n = n_0(1+\alpha)$

$$x(\text{N}_2\text{O}_4) = \frac{n_0(1-\alpha)}{n_0(1+\alpha)} = \frac{1-\alpha}{1+\alpha}$$

$$x(\text{NO}_2) = \frac{2\alpha n_0}{n_0(1+\alpha)} = \frac{2\alpha}{1+\alpha}$$

$$K_x = \frac{4\alpha^2}{1-\alpha^2} = 0,68$$

$$p = 1\text{bar}$$

$$K_x = K_p \cdot 1^{-1} = 0,68$$

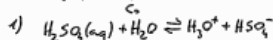
$$p = 10\text{bar}$$

$$K_x = K_p (10)^{-1} = 0,068 = \frac{4\alpha^2}{1-\alpha^2}$$

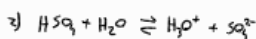
$$\rightarrow \alpha = 0,129$$

Izračunaj pH $0,1\text{M}$ raztopine $\text{H}_2\text{SO}_4(\text{aq})$.

$$\alpha_1 = \frac{c_1}{c_0}$$



$$K_{A1} = \frac{a(\text{H}_3\text{O}^+) a(\text{HSO}_4^-)}{a(\text{H}_2\text{SO}_4) a(\text{H}_2\text{O})} \rightarrow K_1 = \frac{[\text{H}_3\text{O}^+][\text{HSO}_4^-]}{[\text{H}_2\text{SO}_4]} = 10^5 \quad \textcircled{1}$$



$$K_{A2} = \frac{a(\text{H}_3\text{O}^+) a(\text{SO}_4^{2-})}{a(\text{HSO}_4^-) a(\text{H}_2\text{O})} \rightarrow K_2 = \frac{[\text{H}_3\text{O}^+][\text{SO}_4^{2-}]}{[\text{HSO}_4^-]} = 0,012 \quad \textcircled{2}$$

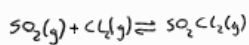
$$\textcircled{3} c_0 = [\text{SO}_4^{2-}] + [\text{HSO}_4^-] + [\text{H}_2\text{SO}_4]$$

$\textcircled{4}$ elektroneutralnost

$$[\text{H}_3\text{O}^+] = [\text{HSO}_4^-] + 2[\text{SO}_4^{2-}]$$

$$\rightarrow [\text{H}_3\text{O}^+] \approx 0,12\text{M}$$

$$\text{pH} = -\log_{10} a(\text{H}_3\text{O}^+) \approx -\log_{10} [\text{H}_3\text{O}^+]$$



$$T = 102,1^\circ\text{C} = 375,25\text{K}$$

$$p^\circ(\text{SO}_2) = 335,9\text{ mmHg}$$

$$p^\circ(\text{Cl}_2) = 358,8\text{ mmHg}$$

$$p = 695,8\text{ mmHg}$$

↳ ravnotežni

$$K_p, \Delta_r G^\ominus, \alpha(\text{SO}_2\text{Cl}_2) \text{ pri } p = 2\text{ bar}$$

$$\Delta_r G^\ominus = -RT \ln K_a$$

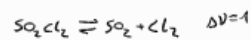
$$101,3\text{ kPa} = 760\text{ mmHg}, 1\text{ bar} = 750\text{ mmHg}$$

	SO ₂	Cl ₂	SO ₂ Cl ₂
zadržetak	335,9	358,8	0
zreagiralo	-p ₃	-p ₃	p ₃
ravnotežje	335,9-p ₃	358,8-p ₃	p ₃

$$p = 335,9 - p_3 + \dots = 335,9 + 358,8 - p_3 \rightarrow p_3 = 48,9\text{ mmHg}$$

$$K_p = \frac{\frac{p(\text{SO}_2\text{Cl}_2)}{p^\circ}}{\frac{p(\text{SO}_2)}{p^\circ} \frac{p(\text{Cl}_2)}{p^\circ}} = 0,412$$

$$b) p = 2\text{ bar} \quad K_p = 0,412^{-1}$$



$$n_0 \quad 0 \quad 0$$

$$-\alpha n_0 \quad \alpha n_0 \quad \alpha n_0$$

$$n_0(1-\alpha) \quad \alpha n_0 \quad \alpha n_0 \quad n = n_0(1+\alpha)$$

$$K_x = \frac{\alpha^2}{1-\alpha^2} = K_p \left(\frac{p}{p^\circ}\right)^{-\Delta V} = \frac{1}{2} K_p$$

$$\alpha = \sqrt{\frac{\frac{1}{2} K_p}{1 + \frac{1}{2} K_p}} = \dots$$

$$x(\text{SO}_2) = \frac{\alpha}{1+\alpha} = x(\text{Cl}_2)$$

$$x(\text{SO}_2\text{Cl}_2) = \frac{1-\alpha}{1+\alpha}$$